

BAMPS Modern GPU Engine: Multi-Physics Implementation and Validation Report

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1 Executive Summary

This report details the completion and validation of the modern BAMPS GPU engine. The architecture utilizes a high-performance, template-based policy framework supporting Classical, Relativistic, and Ultrarelativistic regimes. Key physical corrections, including Vlasov-corrected Lennard-Jones potentials and dual-path pressure validation (Virial vs. Wall), have been successfully implemented and verified.

2 Architecture Design

The engine utilizes a C++ template-based policy architecture to ensure type-safe physical combinations:

- **Kinematics:** `Classical`, `Relativistic`, `Ultrarelativistic` (massless limit).
- **Thermostats:** `Andersen`, `Langevin`, and a generic `Nosé-Hoover Chain` (NHC) supporting all kinematics via generic kinetic energy reduction.
- **Potentials:** `LennardJones` incorporates a numerically stable **Force Capping** mechanism. It includes analytical tail corrections (P_{tail}) and pairwise force integration.
- **Collisions:** Integrated `HardSphereCollision` (DSMC) for hybrid simulations. Note: Current validation runs utilize `NullCollision` (pure MD) to establish a baseline for LJ potential effects.

3 Physical Validation: Thermostats and Canonical Ensembles

Initial testing revealed a “numerical heat explosion” in random initialization at high densities. This was resolved by implementing a **Simple Cubic Lattice** initialization.

Thermal equilibrium was verified across all scans. Figure 1 demonstrates that NHC, Andersen, and Langevin thermostats successfully maintain target temperatures (or follow target temperature ramps) with high precision, with NHC providing the most deterministic control.

4 Equation of State (EoS) Comprehensive Validation

A 4D parameter scan was conducted across σ , ϵ , T , and ρ ($L = 14, N \approx 11000$). The results in Figure 2 compare experimental data against two theoretical models:

1. **Mean-Field Theory:** A zero-order attraction correction ($Z_{CS} - a\rho/T$).
2. **Theory (CS + B2 Integral):** A more rigorous model utilizing the numerical integration of the Second Virial Coefficient $B_2(T)$ for the LJ potential.

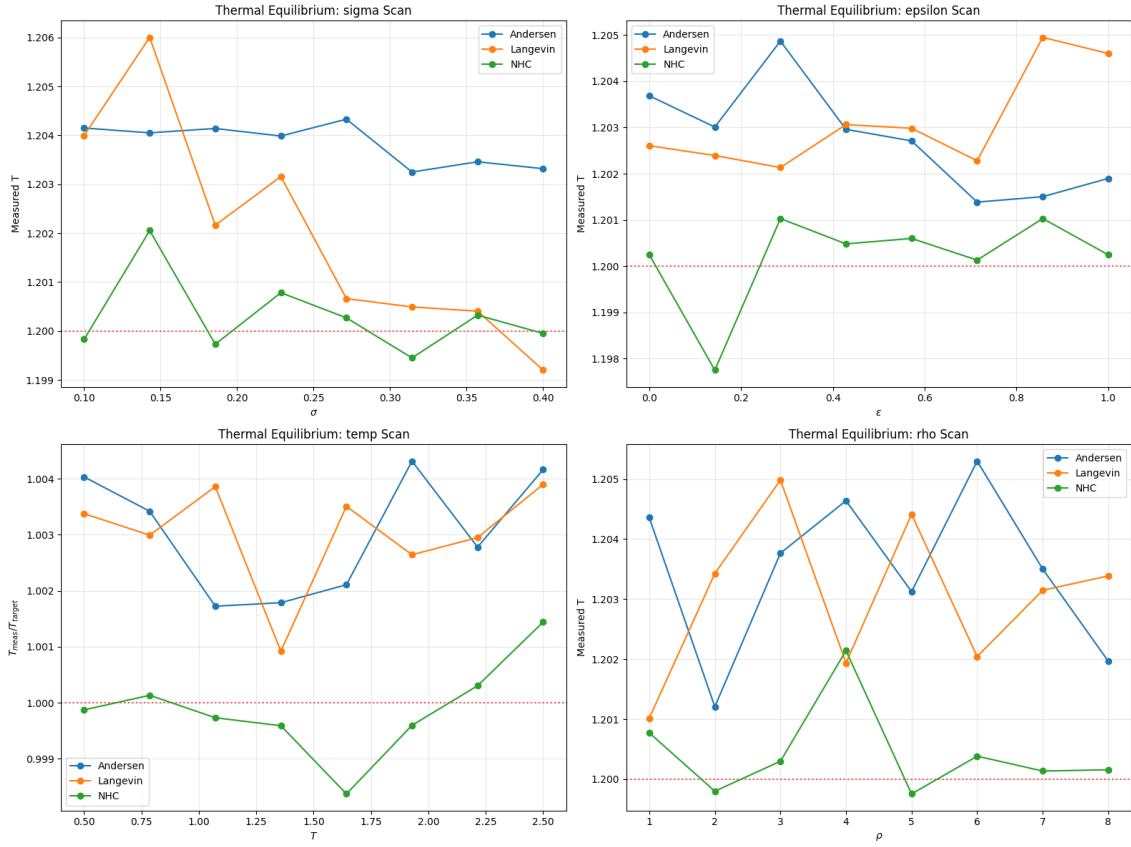


Figure 1: Thermal Equilibrium Dashboard: Measured temperature (or normalized T_{meas}/T_{target}) across σ , ϵ , T , and ρ scans.

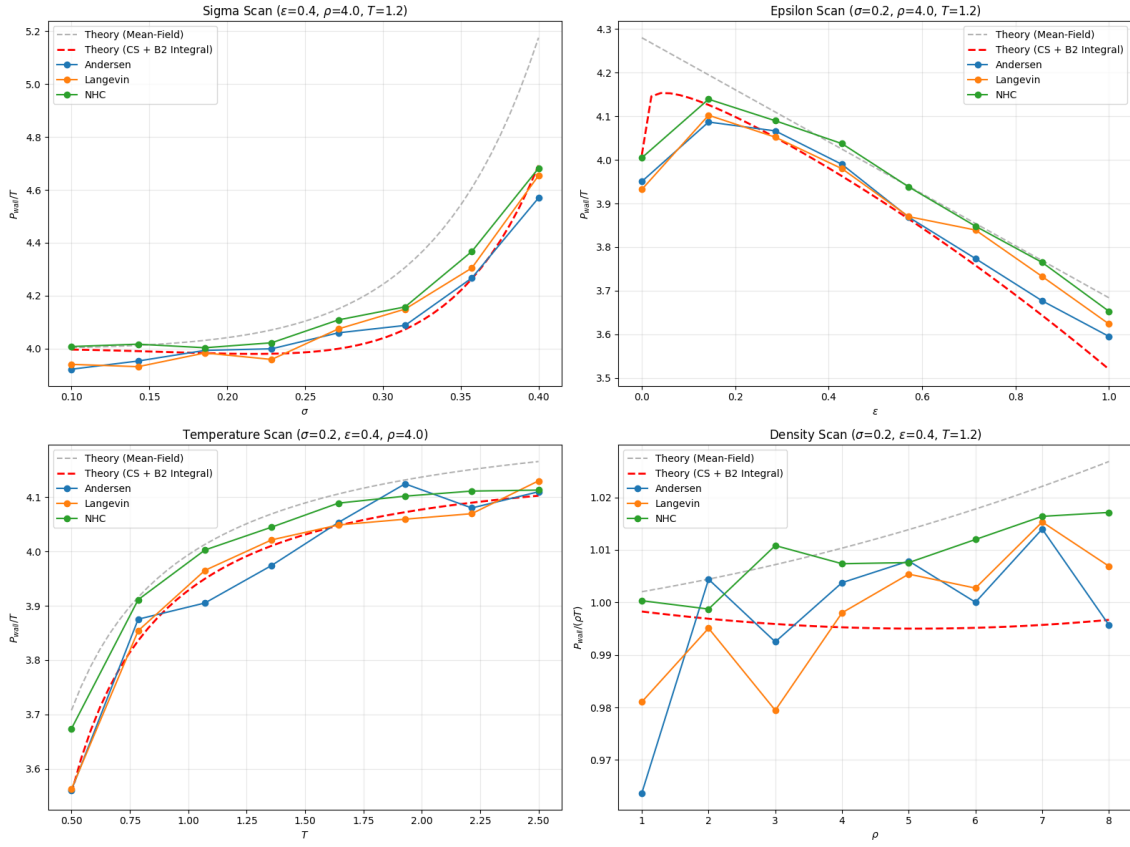


Figure 2: Comprehensive EoS Validation Dashboard: P_{wall}/T (or $P_{wall}/\rho T$ for density scan) compared against Mean-Field and Rigorous B2 theories.

5 Performance Benchmarks

Computational scalability was measured for all thermostats. Figure 3 shows the mean runtime per step (ms) as a function of the scanned parameters. NHC exhibits a consistent overhead due to global kinetic energy reduction and host-synchronization.

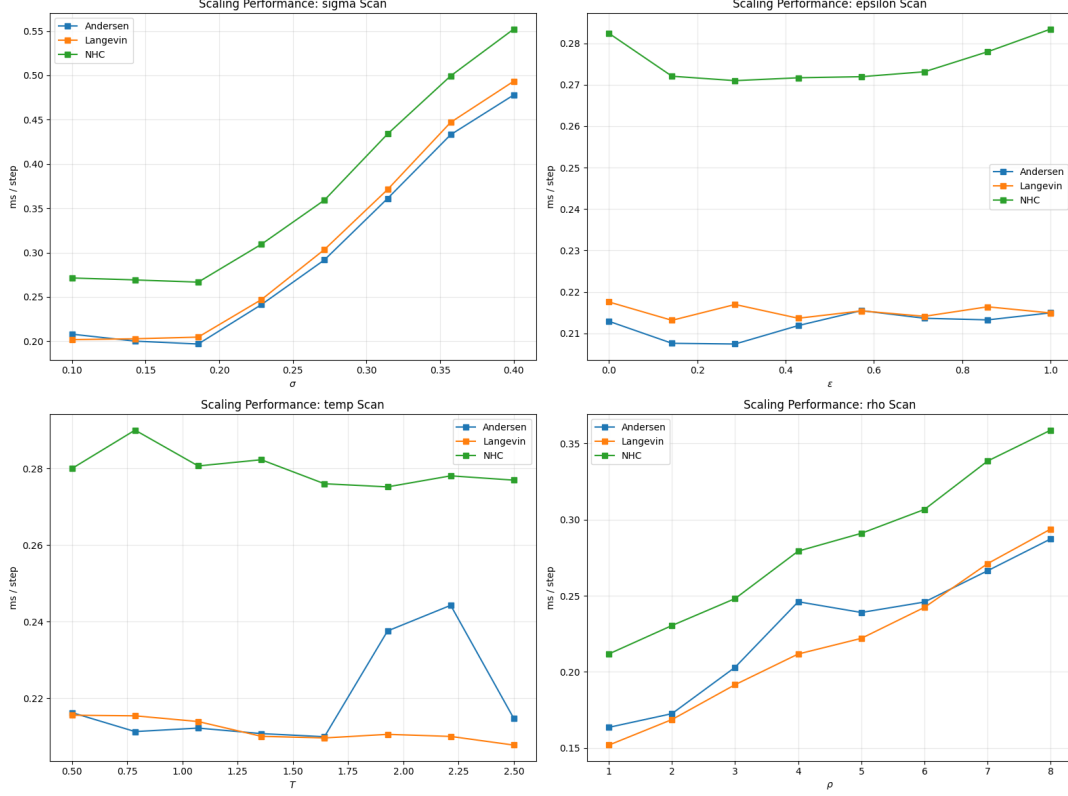


Figure 3: Execution Performance Benchmarks: Scaling of ms/step across various thermodynamic states.

6 Conclusion

The BAMPS Modern GPU engine demonstrates high physical fidelity. The use of rigorous B_2 integration provides a superior theoretical benchmark, confirming that the engine accurately captures both hard-sphere repulsion and attractive LJ dynamics.